Operations on sets

1

a The set of all the elements in the universal set that are not in set A.

b The set containing all the elements that are members of set A or of set B or of both sets.

c The set containing all the elements that are common to both set A and set B.

d The set of elements that belong to set A, but not to set B.

2 $\{3, 7\}$

3

a $\{0, 4, 16, 36, 64\}$

b $\{1, 3, 5, 7, 15, 21, 35, 105\}$

c $\{5, 10, 15, 20, 25, 30, 35, 40, 45\}$

4

a $\{1, 2, 3, 4, 5, 6\}$

b $\{1, 2, 3, 4, 5, 6, 7, 8\}$

c $\{1, 2, 3, 4, 5, 6\}$

d $\{1, 2, 3, 4, 5, 6, 7, 8\}$

5

a

i $\{1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36\}$

ii $\{1, 2, 3, 4, 6, 12\}$

b

i $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 17, 19, 23, 29, 31, 37\}$

ii $\{2, 3, 5, 7, 11\}$

6

a

i The universal set.

ii The universal set, except for the elements of A and the elements of B.

b

i The universal set.

ii The universal set, except for the elements of A and the elements of B.

7

a The universal set, except for the elements that are in both A and B.

b The set of common multiples of 4 and 6 greater than or equal to 70.

c No

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8

a The set of integers that are not multiples of 4 less than 70 or multiples of 6 less than 70.

b The set of integers that are not less than 70, and are multiples of both 4 and 6.

c No

9

a $\{n, p, m\}$

b $\{f, a\}$

10

a $\{85, 20\}$

b $\{19, 69, 17, 35, 29, 85, 20\}$

Complementary events

11

a Not complementary

b Complementary

c Not complementary

d Not complementary

12

a $\frac{1}{5}$

b 0.36

13 0.44

14

a $\frac{34}{69}$

b $\frac{35}{69}$

15

a $\frac{37}{139}$

b $\frac{60}{139}$

c $\frac{102}{139}$

d $\frac{79}{139}$

16

a Picking a number less than or equal to 61.

b $\frac{39}{100}$

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17

a $\frac{5}{6}$ b $\frac{2}{3}$ c $\frac{1}{2}$ d 1

e 0

18

a $\frac{25}{26}$ b $\frac{23}{26}$ c $\frac{17}{26}$ d $\frac{10}{13}$

e $\frac{21}{26}$

19

a $\frac{1}{4}$ b $\frac{1}{4}$ c $\frac{3}{4}$

20

a $\frac{1}{13}$ b $\frac{1}{13}$ c $\frac{12}{13}$ d $\frac{1}{2}$

e $\frac{1}{2}$

21

a 66 b 20

22

a $\frac{19}{30}$ b $\frac{9}{10}$

23

a $\frac{3}{13}$ b $\frac{1}{26}$ c $\frac{2}{13}$ d $\frac{1}{26}$

e $\frac{25}{26}$ f $\frac{51}{52}$ g $\frac{5}{13}$ h $\frac{1}{52}$

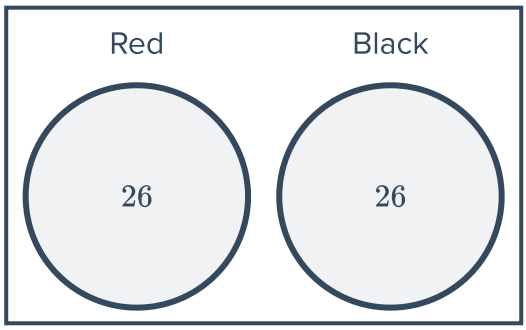
i $\frac{12}{13}$

24 $\frac{5}{36}$

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Probabilities from Venn diagrams

25



26 **a** $\frac{30}{67}$ **b** $\frac{32}{67}$ **c** $\frac{37}{67}$ **d** $\frac{35}{67}$ **e** $\frac{19}{67}$

27 **a** $\frac{2}{9}$ **b** $\frac{7}{54}$ **c** $\frac{19}{27}$ **d** $\frac{8}{27}$

28 **a** **i** Yes **ii** No **iii** No **iv** No **b** $\frac{9}{20}$

29 **a** $\frac{16}{33}$ **b** $\frac{7}{33}$ **c** $\frac{38}{99}$ **d** $\frac{89}{99}$

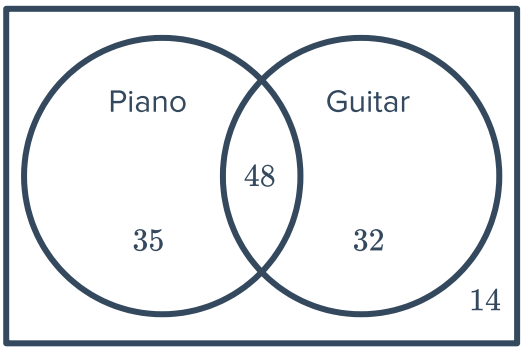
30 **a** $\frac{145}{224}$ **b** 0 **c** $\frac{73}{224}$ **d** $\frac{109}{112}$ **e** $\frac{3}{112}$

31 **a** $\frac{314}{535}$ **b** $\frac{246}{535}$ **c** $\frac{68}{535}$ **d** $\frac{499}{535}$ **e** $\frac{36}{535}$ **f** $\frac{37}{107}$

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32

a



b i $\frac{16}{43}$ ii $\frac{115}{129}$ iii $\frac{14}{129}$

33

a 18 **b** $\frac{93}{172}$ **c** $\frac{93}{172}$

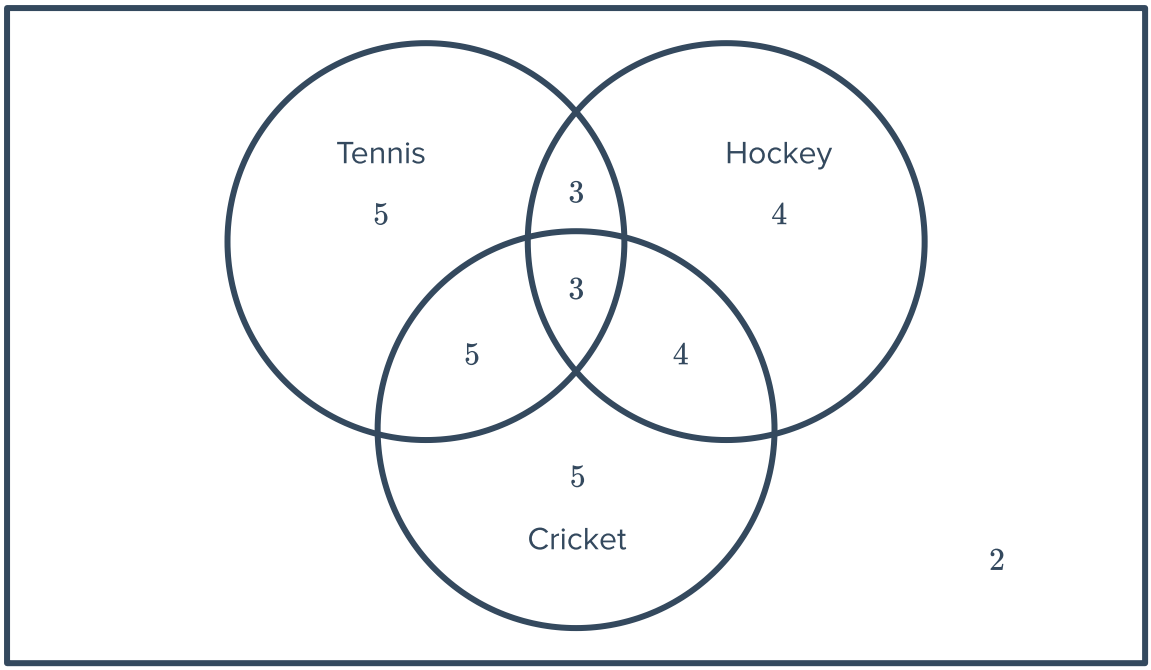
34

a $\frac{565}{1232}$ **b** $\frac{235}{1232}$ **c** $\frac{923}{1232}$ **d** $\frac{131}{1232}$

e $\frac{13}{154}$ **f** $\frac{93}{616}$ **g** $\frac{159}{176}$

35

a



b $\frac{3}{31}$

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